

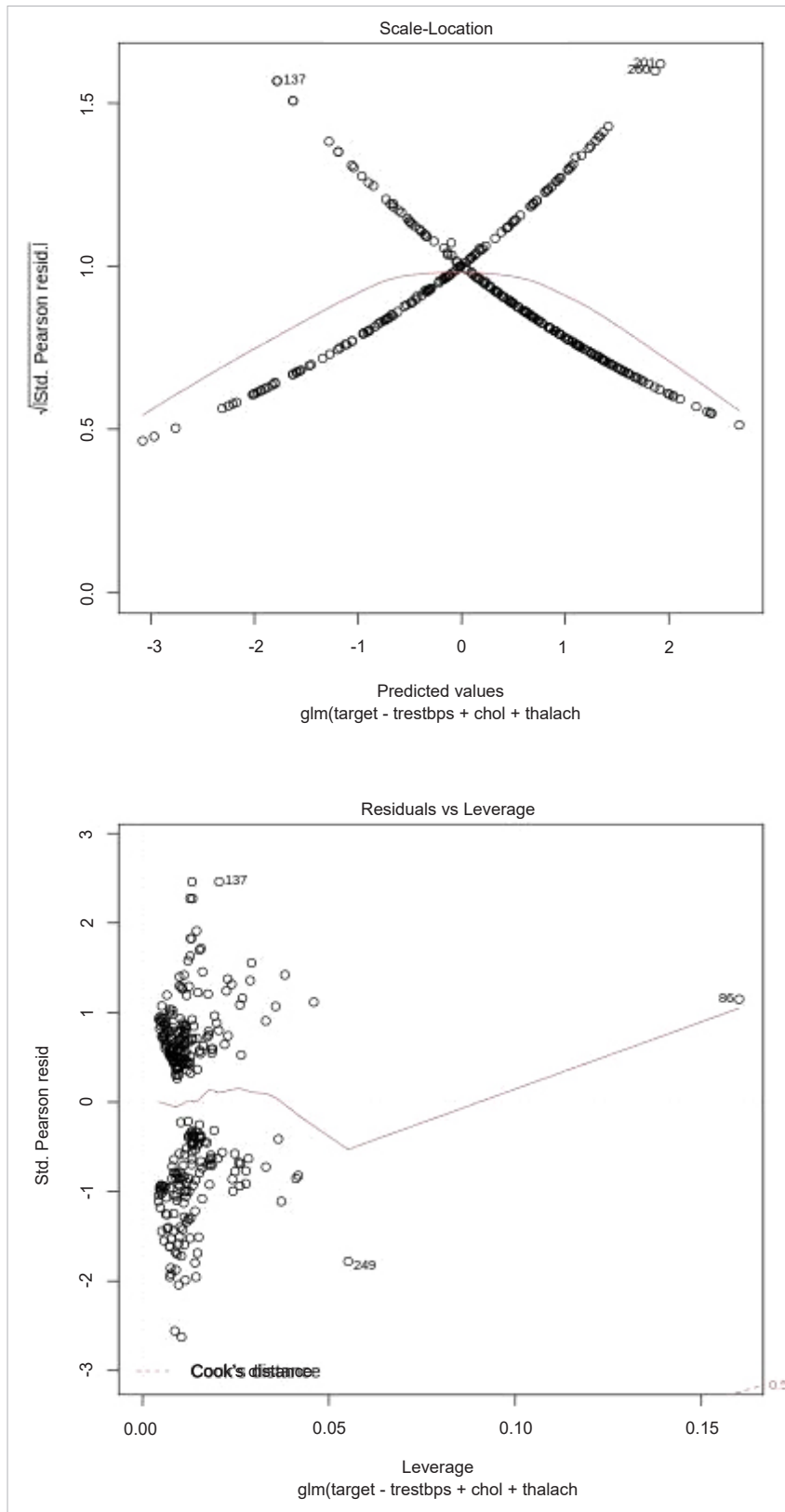


Fig. 8: Graphical representation of residuals vs leverage

for predictive healthcare analytics. The GLM, as implemented in R, offers a robust foundation for predicting heart attack probabilities based on a variety of input features. Its flexibility in handling different types of response variables and accommodating complex relationships between predictors enhances the accuracy of risk predictions.

The integration of random forest in the healthcare application further enhances this accuracy. Random forest leverages multiple decision trees, leading to a more robust and accurate prediction model. It can handle high-dimensional data and capture intricate patterns in the dataset.

Similarly, the inclusion of decision trees provides interpretable and actionable insights into the decision-making process. Decision trees, when used in conjunction with IoT data, allow for the identification of critical features influencing heart attack risks. The simplicity and transparency of decision trees facilitate the communication of results to both healthcare professionals and individuals.

In IoT-based healthcare, the collaborative use of GLMs, random forests, and decision trees in R synergises the strengths of these models to create a comprehensive predictive framework. The combination of these techniques not only enhances accuracy but also provides a holistic view of the potential risk factors associated with heart attacks. **EFY**

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The author, Dr Maheswari R., is Dean Innovation, Sri Eshwar College of Engineering, Coimbatore, Tamilnadu. She has work experience of more than 22 years in industry and academics, has published eight patents, and is the recipient of many awards.